NIHAD ASFIRANE

MSc AI & Machine Learning · MSc Power Engineering

Liverpool, UK | +44 7506 992 478 | nihadasfirane2001@gmail.com
GitHub | Portfolio | LinkedIn

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PROFILE

Engineering graduate with an MSc in Artificial Intelligence & Machine Learning (Distinction) from Liverpool John Moores University and an MSc in Power Engineering. Experienced across the full machine learning workflow, from data preparation and model training to containerised deployment, through projects in computer vision, NLP and GPU-accelerated ML. My dissertation (81%) applied deep reinforcement learning to control a grid-tied inverter. Seeking roles applying machine learning and deep learning to electrical and power engineering.

TECHNICAL SKILLS

PROGRAMMING
- ▸ Python
- ▸ C
- ▸ MATLAB
- ▸ VHDL

ML / AI & DATA
- ▸ Machine Learning
- ▸ Deep Learning
- ▸ TensorFlow / PyTorch
- ▸ Scikit-learn, Pandas, NumPy
- ▸ MLOps

TOOLS & HARDWARE
- ▸ Simulink
- ▸ Proteus & Multisim
- ▸ TI DSP (LAUNCHXL-F28379D)
- ▸ Arduino / Embedded
- ▸ Jupyter Notebook / VS Code
- ▸ LaTeX

EDUCATION & QUALIFICATIONS

MSc Artificial Intelligence (Machine Learning) · *Liverpool John Moores University, UK* — *2025 – 2026*

Distinction (73%) · Modules: Deep Learning (75%), Enterprise ML (76%), Accelerated ML (72%), Foundations of ML (68%), Advanced Topics in Deep Learning (63%)

- ▸ Advanced study of deep learning, computer vision, NLP, GPU-accelerated machine learning and model deployment.

MSc Power Engineering · *IGEE – University of Boumerdes, Algeria* — *2023 – 2025*

- ▸ Advanced study of power systems, power electronics, electric machines, protection and renewable energy integration.

BEng Electrical Engineering & Electronics · *IGEE – University of Boumerdes, Algeria* — *2019 – 2023*

- ▸ Foundational study of circuits, electronics, digital systems, instrumentation, control systems and computer architecture.

KEY PROJECTS

Deep RL Control of a Grid-Tied Inverter · *MSc Dissertation (81%) – LJMU* — *2026*

- ▸ DQN agent directly switching a single-phase H-bridge inverter in Simulink via the MATLAB Engine (5.45% current THD).

Object Detection for Sub-Saharan Wildlife Camera Trap Images + Deployment (MLOps) · *Deep Learning & Enterprise ML – LJMU* — *2026*

- ▸ Faster R-CNN and SSD detectors (72.7% mAP), deployed as a web app with TensorFlow Serving, Flask and Docker.

A BERT-Based QA System for Pressure Ulcer Management and Healthcare Practitioner Training · *Advanced Topics in Deep Learning – LJMU* — *2026*

- ▸ DistilBERT fine-tuned on a custom clinical question-answering dataset of 4,727 questions.

Higgs Boson Classification using Accelerated Machine Learning · *Accelerated ML – LJMU* — *2026*

- ▸ GPU-accelerated XGBoost, Logistic Regression and Linear SVM on 11 million events with NVIDIA RAPIDS (ROC-AUC 0.777).

Skeletal-Based Bird Species Classification using ML Algorithms · *Foundations of ML – LJMU* — *2025*

- ▸ Random Forest, SVM and KNN compared on skeletal bone measurements (best: SVM, 89% accuracy).

Design and Simulation of a Grid-Tied PV Inverter · *MSc Power Engineering Final Year Project – CDER* — *2025*

- ▸ Grid-tied photovoltaic inverter modelled and simulated in MATLAB/Simulink.

Design and Implementation of an NiMH Battery Charger · *BEng Final Year Project – IGEE* — *2023*

- ▸ NiMH battery charger circuit designed and built.

WORK EXPERIENCE

Safety Steward (Part-Time) · *FGH Security Ltd, UK* *Oct 2025 – Present*

- Crowd safety, compliance and customer support at large events, including Anfield Stadium and Chester Racecourse.

Head of Sponsorship Department · *IEEE Student Branch – University of Boumerdes* *Sep 2022 – Jul 2023*

- Led the sponsorship team, developed corporate partnerships, and secured funding for multiple technical student events.

Sponsorship Manager · *IEEE Student Branch – University of Boumerdes* *Sep 2021 – Jul 2022*

- Negotiated with sponsors to secure funding for student engineering events and competitions.

TRAINING & ADDITIONAL EXPERIENCE

- Renewable Energy Systems Training · **CDER**, Algeria (2025)
- Energy Efficiency & Optimisation Training · **APRUE**, **Ministry of Energy and Mines**, Algeria (2024)
- Power Plant Operations Training (LM2500 Gas Turbines) · **SPE, Hamma Power Plant**, Algeria (2023)
- Industrial Electrical Systems Training · **GICA Cement Plant**, **Meftah**, Algeria (2021)

SOFT SKILLS

- **Teamwork & leadership:** led the IEEE sponsorship team and worked in group projects.
- **Communication:** with sponsors, stakeholders and the public at large-scale events.
- **Time management:** balanced full-time MSc study with part-time work.
- **Problem-solving:** applied across engineering and machine learning projects.
- **Adaptability:** studied and worked in Algeria and the UK, in Arabic, English and French.

LANGUAGES

Arabic (Native) **English** (Fluent) **French** (Fluent)

INTERESTS

Music · Sports · Reading · Travelling & Discovering Cultures · Photography · Cooking

REFERENCES

Denis Reilly · *MSc Dissertation Supervisor*
School of Computer Science and Mathematics, Liverpool John Moores University
D.Reilly@ljmu.ac.uk · 0151 231 2279